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Nestlé Line 3 – KitKat 2F Mini

Vision Inspection & Rejection System

End-to-End Solution Design

Phase V1 (Wafer Detection & Rejection) · V2.0 (Top Fill Reveal) · PHASE V2.1 (Finished Bar Reveal)

Field	Details
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1. Project Overview & Process Map

The Defect Bar Rejection System on Nestlé Moulding Line 3 (Usgao, Goa) is designed to achieve individual KitKat bar-level rejection using a precision pneumatic airknife, eliminating the current practice of full-row discards when a single bar is defective. This document describes the complete end-to-end solution covering Phase V1 (wafer detection and tracking), Phase V2.0 (top chocolate fill surface inspection), and Phase PHASE V2.1 (finished bar bottom/logo inspection).

Reference Documents: York Confectionery Quality Control System Manual v1.0 (baseline system), Nestlé Goa URS – Bar Rejection Vision System R7 (target specification). The York system serves as the engineering reference; ***Nestlé Goa modifications are explicitly called out throughout.***

1.1 25-Metre Process Timeline

Position	Mark	Stage	System Active
Wafer Placement	0.0 m	Phase V1 – Wafer Scan (SICK Ruler-3000 3D)	Wafer Detection + QR Read (Entry)
After Backing Pour + Cooling Tunnel	12.5 m	Phase V2.0 – Top Fill Reveal (Baumer Camera)	Surface Inspection (Top Backing Chocolate)
After Demoulding	24.6 m	Phase PHASE V2.1 – Finished Bar Reveal (Baumer Camera)	Surface Inspection (Bottom/Logo) + Rejection Trigger
Airknife Rejection Gate	25.0 m	48-Nozzle Pneumatic Airknife	Selective Bar Ejection

1.2 Production Rate Parameters

Parameter	Value	Notes
Conveyor Belt Speed	0.2 m/s (200 mm/s)	Constant – accumulators maintain this speed
Production Rate	135 rows/minute	45 moulds/min × 3 rows/mould (2F Mini: 2 rows/mould)
Row Interval	~444 ms	Time between successive rows at 0.2 m/s, 88.89mm pitch
Mould Width (inspection)	922 mm	Active chocolate cavity area
Physical Belt Width	1078 mm	Total clearance including mould frame
Fingers per Mould Row	48 fingers	1 per nozzle; 2 fingers = 1 KitKat 2F Mini bar

2. Mould Architecture & Post-Demould Bar Matrix

2.1 2F Mini Mould Cavity Layout

The KitKat 2F Mini mould used at Nestlé Goa (Line 3) contains **48 fingers arranged in 2 rows**:

- **Row 1 (Bottom of Mould):** 24 bars × 2 fingers each = 48 finger slots in the lower pocket row. This row is the leading edge in the direction of mould travel.
- **Row 2 (Top of Mould):** 24 bars × 2 fingers each = 48 finger slots in the upper pocket row. This row is the trailing edge.
- **Finger Pitch:** 19.2 mm centre-to-centre across the 922 mm mould width.
- **Mould Manufacturer:** Agathon (Bottrop, Germany).

IMPORTANT DIFFERENCE FROM YORK: The York system (KitKat Chunky) uses 27 nozzles — one per cavity (27 bars per row, single row per mould). The Goa 2F Mini system uses 48 nozzles — one per finger (24 bars × 2 fingers per bar, 2 rows per mould). This fundamentally changes the nozzle actuation logic from a single-fire per mould to a dual-fire architecture.

2.2 Post-Demould Bar Orientation & Row Inversion

When the mould reaches the demoulding station (25.0m), it undergoes a **180° rotation flip** as part of the standard demoulding drum mechanism. This inversion has critical implications for the airknife firing logic:

- **Before Flip (in mould):** Row 1 is at the bottom of the mould. Row 2 is at the top of the mould.
- **After Flip (on conveyor):** Both rows are now face-up (logo/finished surface facing upward toward the camera). Row 1 (originally at the bottom of the mould) becomes the **leading row on the conveyor** and reaches the airknife gap first.
- **Sequential Arrival at Airknife:** Row 1 (originally bottom) arrives at the 25.0m airknife gap. Approximately 444 ms later, Row 2 (originally top) arrives at the same gap.

NOTE ON ROW LABELLING CONVENTION: Throughout this document, 'Row 1' refers to the mould's original bottom row (leading edge in travel direction, arrives first at airknife). 'Row 2' refers to the mould's original top row (trailing edge, arrives second at airknife). The SICK Ruler-3000 and tracking system must store defect data under these labels to ensure correct nozzle firing sequence.

2.3 Nozzle-to-Bar Mapping Matrix (2F Mini)

The 48 pneumatic nozzles are fixed at 19.2 mm pitch. For the 2F Mini SKU, software maps each of the 24 bars to a pair of adjacent nozzles:

Bar Index	Nozzle Pair	Finger Slots	Width Position (approx.)
Bar 1	Nozzles {0, 1}	Fingers 0–1	0 – 38.4 mm
Bar 2	Nozzles {2, 3}	Fingers 2–3	38.4 – 76.8 mm
Bar 3	Nozzles {4, 5}	Fingers 4–5	76.8 – 115.2 mm
...	...	...	...
Bar 12	Nozzles {22, 23}	Fingers 22–23	422.4 – 460.8 mm
Bar 13	Nozzles {24, 25}	Fingers 24–25	460.8 – 499.2 mm
...	...	...	...
Bar 24	Nozzles {46, 47}	Fingers 46–47	883.2 – 922 mm

Formula: For Bar index i (0-based), nozzle pair = $\{i \times 2, i \times 2 + 1\}$.

The 48 nozzles fire **twice per mould** — once for Row 1 defects, then ~444 ms later for Row 2 defects. If a bar is defective in Row 1 only, only the Row 1 firing activates its nozzle pair. Rows are tracked and fired independently.

3. Phase V1: Wafer Detection System

Phase V1 is directly derived from the York Confectionery Quality Control System (Peacock system, York factory). The Goa implementation adapts the same software architecture to the 2F Mini mould geometry (48 fingers $\times$ 2 rows vs. York's 27 cavities $\times$ 1 row). The Nestlé URS mandates use of the standard Beckhoff vision code framework provided by Nestlé as the baseline.

3.1 Hardware BOM – Phase V1

Item #	Component	Model / Part Number	Qty	Function
1	3D Laser Profile Camera	SICK Ruler-3000 V3DU3-120RM25A (P/N 1132220)	1	Wafer depth profile scan across full mould width
2	QR / Data-Matrix Code Reader	SICK Lector 61 V2D611D-MLSBE5 (P/N 1115203)	4	2x at Phase V1 entry (inspection station), 2x at V2.0 (tracking re-sync)
3	Incremental Rotary Encoder (Camera)	SICK DBS60E-S4CC05000	2	Conveyor speed measurement – one per encoder splitter
4	Encoder Splitter	SICK (compatible)	1	Distributes encoder signal to camera and tracking controller
5	Diffuse Photoelectric Sensor	SICK WTB12-3P2461S58	5	Mould entry detection / trigger sensors
6	Through-Beam Photoelectric Sensor	SICK WS/WE100-2P1239S01	2	Edge synchronisation at inspection and demould gate
7	Inductive Proximity Sensor (Large)	SICK IME18-08BPSZCOS	1	Chain alignment detection
8	Inductive Proximity Sensor (Small)	SICK IME12-04BPSZCOS	2	Chain alignment detection
9	48-Station Solenoid Valve Manifold	SMC SS5Y5 Series + SY5A00-5U1-NA valves	1	Individual nozzle actuation (48 valves)
10	Pneumatic Reject Nozzles	SMC KN-R01-150	52	48 active + 4 spares; 19.2 mm pitch across 922 mm
11	Air Reservoir Tank	SMC VBAT38S1-V (38-Litre Stainless Steel)	1	Compressed air buffer; food-safe filter-regulator
12	Safety Solenoid Dump Valve	SMC VP544-5DZ1-03F-S1-X538	1	Air dump on E-stop / guarding open
13	Beckhoff Industrial PC	Beckhoff C6030-0080 (Intel i7-11850HE, 16GB RAM)	1	Primary vision + tracking IPC; also runs Phase 2
14	Beckhoff HMI Panel	Beckhoff CP2915-0010 (15" touchscreen)	1	Primary operator touchscreen
15	Rockwell ControlLogix PLC	Allen Bradley 1756-L81E (ControlLogix 5580)	1	Airknife nozzle actuation and line PLC
16	Rockwell PLC I/O Adapter	1734-AENT + 1734-IB8/OB8 + 1734-VHSC24	1 set	Digital I/O and high-speed encoder counter modules
17	Reject Conveyor	Cobalt COB-040-ST (powered reject belt)	1	Receives blown-down defective bars

18	Reject Conveyor VFD	Allen Bradley PowerFlex 525 (25B-A4P8N114)	1	Variable speed drive for reject conveyor
19	Reject Verification Camera (Vision 4)	SICK (Model TBD – 2D Vision Camera)	1	Vision-based verification of rejected bars on belt
20	Hygienic Panel	Rittal HD 1310600 + 1307600	1 set	Enclosure panels
21	Tower Lamp	Patlite LR6-502PJBW-RYBGC	1	Visual status indicator (Good/Bad row)
22	Emergency Stop	Schneider XB5AS8445	2	E-stop buttons at inspection and reject stations

3.2 Camera Placement – SICK Ruler-3000 (Phase V1)

3.2.1 Physical Mounting

- **Position:** 0.0 m mark on the production line, immediately after the wafer-placing machine and before the backing chocolate depositor.
- **Mounting:** Mounting should be done on the gantry where current line scanner cameras are placed. Marking should be done so that IF replaced camera calibration isn't required.
- **Camera Axis:** Laser line must project **perpendicular (90°) to the conveyor travel direction**. Deviation of more than $\pm 1^\circ$ will cause geometric distortion of the 3D depth profile.
- **Nominal Working Distance: 770 mm** above the mould surface. This is the critical dimension: at 700 mm WD the FOV reduces to ~843 mm, cropping the outer finger cavities by ~40 mm on each side.
- **Working Distance Range:** 750 mm to 800 mm acceptable.

3.2.2 mm/Pixel Specification – Phase V1

Parameter	Value	Notes
Sensor	SICK Ruler-3000 V3DU3-120RM25A	Integrated laser line; factory calibrated
Nominal Working Distance	770 mm	Required for 922 mm FOV coverage
Field of View (at 770 mm WD)	922 mm (full mould width)	Covers all 48 fingers
Active Pixels (width)	1,280 pixels	SICK Ruler-3000 native width resolution
Spatial Resolution (mm/pixel)	$922 \div 1280 = 0.72$ mm/pixel (lateral)	Adequate for wafer-scale (>5mm) height profiling
Height Resolution (depth)	<0.1 mm (SICK factory spec)	Sub-mm depth accuracy for wafer thickness detection
Scan Rate	Up to 4,000 profiles/second	Line-triggered by encoder; far exceeds 2.25 rows/sec demand
Min Detectable Wafer Feature	~1.5 mm (lateral)	Wafer thickness detection is depth-based, not lateral

3.3 Encoder Calibration & Speed Verification

Two SICK DBS60E-S4CC05000 encoders are mechanically coupled to the conveyor chain via sprockets with spring-loaded brackets to prevent slip. The York manual uses an identical dual-encoder setup (one for the 3D camera, one for tracking to the reject station).

3.3.1 Encoder Calibration Procedure

- **Step 1 – Physical Marking:** Mark a reference point on a test mould and a corresponding reference mark on the belt frame at the Phase V1 scan line.
- **Step 2 – Known Distance Run:** Advance the conveyor precisely 1,000 mm (measured with a calibrated tape). Record the encoder pulse count from the Beckhoff IPC diagnostic screen.
- **Step 3 – Pulse-per-mm Calculation:** $\text{pulses_per_mm} = \text{recorded_pulse_count} \div 1000$. Store this value in the TwinCAT3 PLC variable 'Encoder_PulsePerMM'.
- **Step 4 – Verification Run:** Advance the conveyor 922 mm (mould width distance). Verify encoder count = $922 \times \text{pulses_per_mm} \pm 5$ pulses. If outside tolerance, re-check sprocket engagement and repeat.
- **Step 5 – Speed Verification:** At nominal 0.2 m/s, the pulse rate should be $200 \times \text{pulses_per_mm}$ pulses per second. Monitor via TwinCAT real-time scope; confirm stability to within $\pm 2\%$.

WARNING: Encoder slip is the primary tracking failure mode. If the spring-loaded mounting bracket loses tension (e.g., after cleaning), positional accuracy degrades and phantom nozzle firings occur. Inspect encoder bracket tension at every cleaning period. Per York manual: lubricate encoder bearing mounts with H1 food-grade grease at every maintenance period (two grease nipples per encoder mount).

3.3.2 Hardware Collaboration: Encoder Math & Scanning Physics

The physical sensors, encoder, and cameras do not operate on standard time-based delays (which fail if the conveyor belt speed fluctuates). Instead, they collaborate using strict encoder pulse mathematics:

- **The Absolute Zero Point:** When the leading edge of a tray breaks the beam of the photoelectric sensor, the Beckhoff IPC establishes a 'Zero Point' and begins counting encoder pulses from 0.
- **3D Line-Scan Physics:** The SICK Ruler-3000 is a line-scan camera. The encoder acts as its heartbeat, sending a direct hardware pulse every fraction of a millimeter (e.g., every 0.5 mm). The camera is forced to take exactly one laser slice per pulse. If the belt speeds up or slows down, the camera slice rate perfectly matches the physical movement, guaranteeing the resulting 3D map is mathematically perfect and undistorted.
- **QR Reader Sync Math:** The QR codes are located in the middle of the left and right sides of the tray (approx. 200 mm from the leading edge of a 400 mm tray). Instead of using a fixed time delay (e.g., 1.0 second), the IPC calculates the precise pulse count. For example, if the encoder is calibrated to 10 pulses per mm, the IPC calculates $10 \text{ pulses} \times 200 \text{ mm} = 2,000 \text{ pulses}$. Exactly at pulse #2,000, the IPC fires the SICK Lector 61 scanners, ensuring the cameras trigger dead-center on the QR code regardless of belt speed.

3.4 Detection Algorithm & Pass/Fail Logic

The Phase V1 algorithm runs on the Beckhoff IPC C6030-0080 within the TwinCAT Vision framework (TF7100 Vision Base + TF7300 Vision Metrology 2D). It directly adapts the York/Peacock inspection algorithm to the 48-finger 2-row Goa mould geometry.

3.4.1 Inspection Pipeline (per Mould)

- **Step 1 – Mould Trigger:** Diffuse photoelectric sensor (SICK WTB12-3P2461S58) detects the leading edge of an incoming mould and sends a digital trigger to the IPC.
- **Step 2 – Encoder-Gated 3D Scan:** The SICK Ruler-3000 begins continuous line-scan acquisition. The encoder provides real-time position data to stitch individual scan lines into a coherent 3D depth map of the entire mould (both rows, all 48 fingers).
- **Step 3 – Depth Thresholding:** Initial depth filter applied. The algorithm establishes the mould cavity baseline (bottom of pocket) and identifies the height range representing the chocolate layer and wafer.
- **Step 4 – Contour Detection:** The algorithm identifies individual cavity contour boundaries for each of the 48 finger slots across 2 rows. **If a cavity boundary cannot be detected** (e.g., chocolate residue bridging cavities), coordinate interpolation is performed from adjacent cavities to ensure all 96 finger positions are always inspected — same behaviour as York.
- **Step 5 – Cavity Analysis per Finger:** For each of the 96 finger slots (48 fingers $\times$ 2 rows), the algorithm measures the wafer height profile and checks against the configured wafer thickness thresholds (minimum and maximum acceptable heights from cavity bottom).
- **Step 6 – Pass/Fail Decision per Bar:** A bar (2 adjacent fingers) passes if BOTH its fingers pass. A bar fails if either finger fails. The fail decision is stored in the defect map as `[Row][Bar_Index] = FAIL`.

3.4.2 Pass / Fail Criteria (Per Nestlé URS – Goa)

Condition	Decision	Action
Full bar – all wafer layers present (100% length)	PASS	Log; no rejection queued
80% or more of wafer length present, all layers intact	PASS	Log; no rejection queued
Full bar – exactly ONE wafer layer delaminated (height reduced, but present)	PASS	Log warning; no rejection
Bar missing MORE THAN 20% of wafer length	FAIL	Queue for rejection at airknife
Bar with TWO or more wafer layers missing (height reduction ≥5 mm)	FAIL	Queue for rejection at airknife
Double-stacked wafer ('sticker') detected – wafer height exceeds maximum	FAIL	Queue for rejection at airknife
>10 FAIL detections within any 1-minute window	ALARM	HMI alarm raised; operator alerted for upstream process fault

YORK REFERENCE: York (Chunky) uses configurable 'Wafer Rejection Threshold' (% of cavity area within acceptable height range). Goa URS specifies 80% minimum wafer length as the pass threshold. The TwinCAT parameter 'Wafer_Rejection_Threshold_Pct' should be initialised to 80.0 and confirmed by Nestlé QA during commissioning. Increasing this value raises the rejection rate; decreasing it reduces sensitivity.

3.5 Tracking Architecture – QR Primary / Encoder Fallback

The tracking system is a **three-layer architecture** providing maximum robustness. This differs from York which uses a two-point QR re-sync (inspection + demould) by adapting to the Goa constraint that QR readers cannot be placed at the vibration-prone PHASE V2.1 demoulding hammer station.

3.5.1 QR Camera Placement (Goa – Adapted from York)

- **QR Reader Pair 1 (Entry):** 2× SICK Lector 61 at the Phase V1 inspection station (0.0 m). One reader per side of the mould, reading the dual data-matrix codes on the mould underside. Records mould ID + defect map into the tracking FIFO database.
- **QR Reader Pair 2 (PHASE V2.0 Re-Sync):** 2× SICK Lector 61 at the V2.0 station (12.5 m). Re-reads the mould ID as the mould passes the top-fill inspection camera. This confirms mould identity mid-line and resets the encoder distance counter for the remaining 12.5 m to the airknife.
- **Departure from York:** York places its 2nd QR pair immediately inside the demould machine. Goa placement can be done at L3 due to vibration from the demoulding hammer and physical space constraints at PHASE V2.1.

3.5.2 Three-Layer Tracking Logic

- **Layer 1 – QR Identity at Entry (0.0 m):** Mould ID read by Lector Pair 1. Defect map stored in FIFO queue: {mould_id, row1_defects[], row2_defects[], entry_encoder_ticks}.
- **Layer 2 – QR Re-Sync at V2.0 (12.5 m):** Lector Pair 2 re-reads mould ID. Software validates that the arriving mould ID matches the next expected mould in the FIFO queue (within a configurable mould count tolerance). On match: encoder tick counter is reset/adjusted. On mismatch: alarm raised; queue flushed for safety.

- **Layer 3 – Encoder Distance + Proximity Trigger at PHASE V2.1 (24.6 m):** From the V2.0 QR re-sync point, the encoder tracks distance to the PHASE V2.1 scan line (12.1 m) and then to the airknife gap (12.5 m). A proximity/photoelectric sensor at the airknife gap provides the FINAL TIMING TRIGGER for nozzle firing, independent of encoder accumulation.

3.5.3 Encoder Shift-Register Fallback Mode

If both QR readers at V2.0 fail to read a code (dirty/damaged), the system automatically falls back to pure encoder shift-register tracking:

- The FIFO queue entry (from Phase V1 QR read) provides the mould ID and defect map.
- The encoder continuously calculates the expected arrival tick at the airknife based on: $\text{target_ticks} = \text{entry_ticks} + (25,000 \text{ mm} \times \text{pulses_per_mm})$.
- The proximity sensor at PHASE V2.1 provides final trigger confirmation.
- HMI displays a 'QR READER DEGRADED – ENCODER MODE ACTIVE' warning.
- Operator can switch modes via the HMI 'Tracking Mode' toggle (QR / Shift-Register).

3.6 Nozzle Actuation Matrix & Dual-Row Firing Logic

3.6.1 Firing Sequence (per Mould)

- **Event 1 – Row 1 Arrival:** The proximity sensor at the airknife gap detects the leading edge of Row 1. The PLC retrieves the Row 1 defect map for the current mould and opens the corresponding nozzle solenoids (nozzle pairs for all defective bars in Row 1) for the configured Reject Spray Time (default: TBD ms, calibrated on-site). Row 1 defective bars are blown off the conveyor.
- **Delay:** Software waits ~444 ms ($88.89 \text{ mm} \div 200 \text{ mm/s}$) for Row 2 to arrive at the gap. The exact timing is encoder-calculated, not fixed.
- **Event 2 – Row 2 Arrival:** The proximity sensor detects Row 2. The PLC retrieves the Row 2 defect map and fires the corresponding nozzle pairs. Row 2 defective bars are blown off.
- **Clean Mould:** If neither row has any defective bars, no nozzles fire. FIFO queue entry is cleared.

3.6.2 Key Timing Parameters (to be calibrated on-site)

Parameter	Description	Default / Notes
Reject Spray Time	Duration (ms) nozzles remain open per bar	Calibrate empirically on-site per York Section 4
Reject Delay Time	Delay (ms) from proximity sensor edge to nozzle open	Adjust for exact bar position over gap
Row Interval Time	Time (ms) between Row 1 and Row 2 nozzle events	~444 ms at 0.2 m/s; encoder-calculated dynamically
Air Pressure	Compressed air supply to manifold	~5 bar; never exceed 7 bar (York spec)
Conveyor Gap Width	Gap between moulding and packaging conveyors	~43 mm (York commissioning default; verify on Goa line)

YORK REFERENCE – 'Max Row Rejects' Parameter: York implements a configurable maximum number of rejects per row. If this count is exceeded, the system rejects the full row from the packaging conveyor rather than individual bars. The Goa URS does NOT explicitly require this feature. However, per York operational precedent, Neebal should verify with Nestlé Goa whether a full-row fallback threshold is required, and if so, agree the threshold value with Nestlé QA during commissioning.

3.7 Reject Verification

- **Through-beam type photoelectric sensor (WS/WE100-2P1239S01):** Verify if fall occurred and another for if rejection bin is full
- **Positive Verification:** If rejection was fired and camera detects a bar within the expected time window: rejection confirmed, logged to HMI.
- **Negative Verification (Miss):** If rejection was fired but NO bar is detected on the reject belt within the window: HMI alarm raised — 'REJECTION MISS – CHECK LINE'. Operator must inspect the gap zone. The URS requires this alarm.
- **False Trigger Guard:** If the rejection camera detects a bar when no rejection was fired: HMI alarm 'UNEXPECTED OBJECT ON REJECT BELT'. Indicates a bar fell off the production belt unexpectedly.
- **Row Misalignment Risk:** Per the Goa URS: the alignment of rows during demoulding must be straight. Misalignment can cause defective bars to pass the nozzle gap without being struck. Gantry design and conveyor alignment must be validated during commissioning.

3.8 Operating Modes & HMI States

Mode #	Mode Name	Description	Rejection Active?
1	IDLE	System powered but not scanning. Awaiting operator start command.	No
2	SCAN ONLY	QR codes being read at entry and V2.0 stations. No wafer inspection or rejection.	No
3	INSPECT	QR scanning + 3D wafer inspection active. Defect results visible on HMI but no nozzle outputs.	No
4	SCAN, INSPECT & REJECT	Full operational mode. All stages active. Defective bars are physically ejected.	Yes
5	E-STOP / SAFE STATE	Emergency state. Compressed air dumped via solenoid dump valve. Reject conveyor stopped. Guarding unlocked.	No – Air Dumped

The reject conveyor (Mode 4) must be manually started from the reject station HMI after switching to full operation mode – it does NOT auto-start on mode change (York SOP, adopted for Goa).

3.9 Stoppages, E-Stop & Restart Procedures

3.9.1 Planned Stoppage (Recipe Change / Cleaning / Maintenance)

- Operator presses 'Request Unlock' on HMI. System completes current mould scan cycle before stopping (does not halt mid-scan).
- Air dump valve vents remaining compressed air. Reject conveyor stops.
- FIFO tracking queue is **saved to persistent memory** with timestamp and last confirmed mould ID.
- Guarding doors unlock. Engineers may access the reject station safely.
- After maintenance: operator confirms FIFO queue status on HMI (Clear Queue / Resume Queue). **Clear Queue is recommended after any belt adjustment or mould changeover.** This prevents phantom rejections based on stale pre-stoppage data.

3.9.2 Unplanned E-Stop

- Air dumped immediately via safety solenoid. System enters Safe State (Mode 5).
- FIFO queue frozen. Last successfully tracked mould ID logged with encoder tick at time of E-stop.
- On restart: operator must review HMI 'FIFO Queue Status' screen. Operator confirms whether to: (a) **Resume** – re-arm the queue from last known mould (only safe if belt did not move during E-stop), or (b) **Clear** – flush the queue and restart clean.
- Default recommendation: **Always Clear Queue on E-stop restart** unless line has been stationary for less than 10 seconds and mould positions are known.

CAUTION: Phantom rejections after restart are one of the highest-risk operational failure modes. If a defect map for a mould already passed through the airknife is still in the FIFO queue when the system restarts, it will fire nozzles at the wrong bars. Always clear the FIFO queue after any unplanned stop unless position can be positively confirmed.

3.10 York-Derived Warnings & Goa Mitigations

#	York Warning	Risk	Goa Mitigation
W-1	Camera/sensor positional marking before replacement	If a camera is replaced without marking its original position, recalibration is required and production is lost	Apply permanent silicone marking dots on the gantry frame at all four mounting feet of the SICK Ruler-3000 and all Lector 61 brackets. Photograph positions before any maintenance. Store reference image in Beckhoff IPC.
W-2	QR reader covers must be cleaned regularly	Chocolate dust accumulating on the reader cover window degrades read rates. At York, polycarbonate covers slide to the side for cleaning during operation.	SICK Lector 61 V2D611D-MLSBE5 (IP65/67 rated) is less susceptible than York's open readers. However, a daily visual inspection and wipe-down protocol with a soft cloth is mandatory. If read failure rate exceeds 5% of moulds in a shift, initiate immediate cleaning.
W-3	Reject conveyor castors must be locked during operation	Vibration can cause the reject conveyor to shift, misaligning the sensor and creating gaps for bars to fall off	Lock all castors before entering Mode 4. Include in pre-shift checklist. Label castor lock points with red safety tape.
W-4	Encoder bearing lubrication (H1 food-grade grease, two grease nipples per mount)	Dry encoder bearings increase friction, causing slip and positional error in tracking	Schedule lubrication at every maintenance period (frequency to be determined on-site from York's experience). Document in the site maintenance log. Use only H1-certified food-grade grease.
W-5	Nozzle beam alignment after any cleaning or belt work	If the nozzle beam slides during cleaning, individual nozzle positions shift, causing misfires (nozzle hits bar edge, not centre)	Mark the beam bolt positions with paint pen before each cleaning shift. After reassembly, verify beam alignment against the reference mark before restarting. Per York: gap between moulding and packaging conveyors must be ~43 mm (verify exact value for Goa line during commissioning).

4. Phase V2.0: Top Fill Reveal (12.5 m Station) · Location R9

4.1 Inspection Target & Camera Geometry

At the 12.5 m mark, the chocolate bars are still inside the mould pockets after the backing chocolate has been deposited and the mould has travelled through the cooling tunnel. The V2.0 camera system inspects the **top surface of the cooled backing chocolate** while the bars are still in mould cavities. **All devices will be placed at R9**

- **Camera Axis:** Perpendicular (straight down, 90° to belt surface). A slight off-axis angle would cause the mould cavity walls to cast shadows on the backing chocolate surface, obscuring defects near the cavity edges.
- **Lighting Geometry:** Off-axis, oblique angle illumination (cross-polarized). The polarizing film on the LED light and the orthogonal analyzing filter on the lens together eliminate specular glare from the shiny cooled chocolate surface, revealing sub-surface defects and fill anomalies.

- **Triggering (The Pulley FIFO Queue):** Because the tray travels vertically down a cooling pulley, an encoder cannot track distance. Instead, a strict FIFO digital queue driven by two photoelectric/proximity sensors is mandatory.
- **Hardware Collaboration:** 1) Top Entrance: A proximity sensor detects the tray entering the cabinet, triggering the SICK Lector QR scanner. The read ID is pushed to a digital queue. 2) Bottom Exit: A second proximity sensor detects the tray reaching the bottom. It triggers the Baumer camera and strobe light. 3) The Link: The Beckhoff IPC assigns the oldest ID in the queue to the newly taken image.
- **Critical Warning (The Blind Zone):** The vertical drop is a 'Blind Zone'. If a tray is manually removed, the physical order changes but the digital queue does not, causing permanent desynchronization. Solution: Magnetic safety switches **MUST** be installed on cabinet doors to flush the queue upon opening.
- **QR Read Failure (Desynchronization Risk):** If the top QR scanner fails to read due to chocolate fouling on the code or the scanner lens, the tray enters the pulley without an ID, throwing the FIFO queue out of sync. Solution: The Beckhoff IPC must track the sequence gap and raise an immediate 'QR SCAN FAILURE - CHECK LENS' alarm on the HMI and line tower light to warn Nestlé operators.
- **Nozzle Execution (The Merge):** The IPC merges V2.0 defects with Phase V1 defects for that Tray ID. The IPC transmits the merged array to the Rockwell PLC. The PLC uses its Shift Register and a final encoder to track the tray to the airknife and fire the nozzles.

4.2 Combined Phase 2 Bill of Materials (PHASE V2.0 & PHASE V2.1 Hardware)

STATUS: Upgraded to a Dual-Camera 24.3 MP Array per stage (4 cameras total) to eliminate optical parallax distortion at the edges of the mould. WARNING: This generates ~100 Megapixels of data per cycle. The Beckhoff IPC MUST be upgraded to a GPU-enabled model with dedicated NICs to meet the 1.5-second latency budget.

Component	Specification	Qty	Notes
Area Scan Camera	Baumer VCXG.2-241M.I (24.3 Megapixel Monochrome)	4	2 for V2.0, 2 for PHASE V2.1. Sony IMX540 CMOS sensor (5328×4608).
Lens	12mm C-mount (Manual Iris with lock)	4	High-resolution low-distortion lens for 1/1.2" sensor.
Polarizing Filter	Linear polarizing filter (C-mount threaded)	4	Mounted on each lens, orthogonal to light polarization.
LED Light Panel	Cross-polarized White LED (min. 1000mm width)	2	1 panel for V2.0, 1 for PHASE V2.1. Essential for glare elimination.
Photoelectric (PE) Sensor	SICK WTB12 (or equivalent 24V PNP)	4	Triggering sensors: 2 for V2.0 Pulley Queue, 2 for PHASE V2.1 Demoulding FIFO.
QR Scanner	SICK Lector 61 V2D611D	4	1 for V2.0 Pulley entrance, 1 for PHASE V2.1 Demoulding entrance (Cabinet L3).
Strobe Controller	Hardware strobe controller	2	Microsecond synchronisation for LED panels and cameras.
Beckhoff IPC	Beckhoff C6043 (or similar) with NVIDIA RTX GPU	2	Replaces standard C6030. Mandatory for processing 4x 24.3MP AI inference within 1.5s. (UPGRADE REQUIRED from Phase 1)
IPC Network Expansion	4-Port Gigabit Ethernet PCIe Card (GigE Vision)	2	Mandatory. Each 24.3MP camera MUST have a dedicated physical NIC port to prevent bandwidth choke.

Dual-Camera 24.3 MP Array Performance (Per Stage)

Parameter	Value	Notes
Field of View (per camera)	~500 mm width	2 cameras side-by-side cover the 1000mm belt.
Spatial Resolution		5328 pixels ÷ 500 mm. Ultra-high precision.
1 mm defect coverage	~11 × 11 pixels	Exceptional clarity for AI/blob 'Warning' threshold.
3 mm defect coverage	~32 × 32 pixels	Absolute certainty for 'Rejection' classification.
Optical Advantage	Parallax minimized	Cameras look straight down into cavities, eliminating blind spots.
Compute Load	49.1 MP total per stage	Requires GPU acceleration and 4x dedicated GigE lines.

4.3 Detection Algorithm – Phase V2.0 (Top Fill Surface)

V2.0 inspects the top backing chocolate surface for defects that indicate fill quality issues (under-fill, bubble formation, surface cracks).

4.3.1 Algorithm Architecture

- **Stage A – Classical Image Processing (1–3 mm minor defects):** Thresholding + connected-component blob analysis + contour detection. Identifies and measures surface anomalies. Anomalies in the 1–3 mm range are logged to the HMI as warnings (no rejection trigger).
- **Stage B – Faclon Surface AI Detection model (>3 mm major defects):** Faclon Surface AI Detection model to be run on the Beckhoff IPC NVIDIA GPU. Detects air bubbles >3 mm diameter, large cracks, and exposed base wafers >10 mm². I
- **Decision Merge:** Outputs from Stage A and Stage B are merged. If either stage flags a bar as FAIL (major defect), the bar index and row index are added to the Phase 2 reject map.
- **Phase 2 Reject Map Merge with Phase V1:** The Phase 2 reject map is OR-merged with the Phase V1 defect map. Any bar failing either Phase V1 OR Phase 2 criteria is rejected at the airknife.

4.3.2 Phase 2 Defect Training Data Requirement

CRITICAL DEPENDENCY: The Faclon Surface AI Detection model for Phase 2 CANNOT be trained before cameras are installed on the production line. There is NO existing dataset. The following protocol must be executed during commissioning: 1. Cameras installed and triggered in IMAGE-CAPTURE-ONLY mode (no inference, no rejection). 2. System runs for 1–2 weeks capturing images of every mould pass. 3. Images are transferred to Faclon Labs via SECURE encrypted transfer (VPN tunnel or SFTP – method to be agreed with Neebal and Nestlé IT before commissioning). 4. Nestlé QA team validates and labels defect images before training begins. 5. Faclon trains and validates the model. Deploys back to Beckhoff IPC via Neebal. 6. Phase 2 live rejection activated only after model validation sign-off by Nestlé QA. Estimated timeline from camera installation to Phase 2 live rejection: 4–8 weeks.

5. Phase PHASE V2.1: Finished Bar Reveal (24.6 m Station) · Location L3 – L1

5.1 Inspection Target & Camera Geometry

At the 24.6 m mark, the bars have been demoulded (mould flipped 180°) and are travelling face-up on the packaging conveyor. PHASE V2.1 inspects the **finished chocolate surface and logo embossment** as bars approach the 25.0 m airknife gate.

- **Camera Axis:** Straight down, perpendicular to belt. Both rows of bars (Row 1 leading, Row 2 trailing) are face-up at this point.
- **Inspection Window:** The PHASE V2.1 scan captures an image of the approaching row. The encoder tracks position from the PHASE V2.1 scan line (24.6 m) to the airknife gap (25.0 m) — a distance of 400 mm.
- **Lighting:** Same cross-polarized LED panel approach as V2.0. Critical for revealing logo clarity, surface cracks, and exposed wafer edges on the finished bar face.
- **Critical Warning – Row Skew (The Crooked Drop):** When the mould flips, bars drop onto the belt. If they drop unevenly, the row sits diagonally (skewed) across the belt. Since the pneumatic airknife is perfectly straight, a skewed row makes precision single-bar rejection physically impossible (nozzles may hit adjacent good bars). Mitigation: The PHASE V2.1 Baumer camera must calculate the skew angle. If it exceeds a tolerance limit (e.g., >5 degrees), the IPC instructs the PLC to execute a Full-Row Reject fallback.

- Hardware Collaboration (Demoulding FIFO):** Because the chocolate is demoulded from the tray at cabinet L3, the physical chocolate on the packaging belt no longer has a QR code to scan. To synchronize the data across the flip: 1) A QR scanner and PE sensor placed BEFORE the demoulding flip reads the mould ID **at L3** and pushes the Phase V1 & V2.0 defect array into a 'Demoulding FIFO Queue'. 2) A second PE sensor at 24.6m (**AFTER the R1**) detects the naked chocolate arriving. This sensor directly hardware-triggers the Baumer camera's digital I/O for zero-latency image capture. 3) Simultaneously, the IPC pops the oldest defect array from the Demoulding FIFO and assigns the newly captured PHASE V2.1 image to it, perfectly syncing all three inspection stages.
- Data Transformation (Lateral Matrix Inversion):** When the IPC pops the defect array from the Demoulding FIFO, it must execute a mathematical coordinate transformation. When the mould flips 180° over the demoulding drum, it lands upside down on a belt moving in the same direction. This means the leading row (Row 0) stays the leading row, but the left side of the mould flips over to become the right side of the belt. The IPC must map the coordinates using Lateral Inversion: $\text{New_Row} = \text{Old_Row}$, $\text{New_Column} = (\text{Total_Columns} - 1) - \text{Old_Column}$. For example, a defect at [Row 0, Column 0] in the mould is mapped to [Row 0, Column 23] on the PLC airknife bitmask. Without this mirrored inversion, the airknife will perfectly reject the exact wrong bar on the opposite side of the belt!

5.2 Rejection Latency Budget

The 400 mm gap between PHASE V2.1 scan line and the airknife is the tightest timing constraint in the entire system:

Budget Item	Allocation	Notes
Total physical travel time (400 mm at 0.2 m/s)	2,000 ms	Hard constraint – cannot be changed
Camera exposure + frame transfer to IPC	≤ 50 ms	Hardware-triggered strobe exposure
Faclon Surface AI Detection model inference	≤ 400 ms	Target; validate during commissioning
Classical image processing pipeline	≤ 100 ms	Thresholding + blob analysis
Phase V1/2 defect map merge + nozzle bitmask generation	≤ 50 ms	In-memory operation
EtherNet/IP CIP Write Tag to Rockwell PLC (Airknife_Nozzle_Array)	≤ 50 ms	Local network; negligible if on same subnet
Total software execution budget		Must complete within 1.5 seconds
Safety margin for pneumatic solenoid actuation + air blast	500 ms	Remaining from 2,000 ms window
If AI inference exceeds 400 ms during commissioning testing, the following mitigations must be applied in order: (1) Reduce input image to inspection ROI only. (2) Reduce model complexity.		

5.3 Unified Reject Decision Merge (Phase V1 + V2.0 + PHASE V2.1)

The Beckhoff IPC maintains a unified reject bitmask per mould per row. The final nozzle command written to the Rockwell PLC tag 'Airknife_Nozzle_Array' is the logical OR of all three inspection stages:

- Phase V1 result:** Row 1 / Row 2 defect flags from wafer inspection [48 bits per row]
- V2.0 result:** Row 1 / Row 2 flags from top fill surface inspection [48 bits per row]
- PHASE V2.1 result:** Row 1 / Row 2 flags from finished bar inspection [48 bits per row]
- Final mask:** `final_mask[row][bit] = Phase1[row][bit] OR V2.0[row][bit] OR PHASE V2.1[row][bit]`

This merged mask is transmitted to the PLC as a 48-bit integer. Nozzle pairs are fired for all set bits. The PLC fires the 48-nozzle array twice (once per row, ~444 ms apart).

6. Network Topology & IP Address Table

NOTE: All IP addresses below are PROVISIONAL. Exact values must be confirmed and assigned during on-site commissioning in coordination with Neebal and Nestlé IT. All devices must reside on the same 192.168.1.x subnet for EtherNet/IP communication.

Device	Role	Protocol
Beckhoff IPC C6030 (Camera/Vision Controller)	Primary IPC – Runs TwinCAT3 Vision + Phase 2	EtherNet/IP, GigE Vision
Beckhoff EtherNet/IP Master Module	PLC comms bridge on Beckhoff	EtherNet/IP
SICK Lector 61 #1 (Phase V1 Entry – Side A)	Mould QR read at inspection station, left side	EtherNet/IP
SICK Lector 61 #2 (Phase V1 Entry – Side B)	Mould QR read at inspection station, right side	EtherNet/IP
SICK Lector 61 #3 (V2.0 Re-Sync – Side A)	Mould QR re-read at V2.0 station, left side	EtherNet/IP
SICK Lector 61 #4 (V2.0 Re-Sync – Side B)	Mould QR re-read at V2.0 station, right side	EtherNet/IP
Rockwell ControlLogix EN2T (Comms Module)	PLC Ethernet/IP Comms Module	EtherNet/IP
HMI – Inspection Station (Beckhoff CP2915)	Operator panel at Phase V1 station	TwinCAT HMI
HMI – Reject Station	Operator panel at airknife station	TwinCAT HMI
SMC 48-Station Solenoid Manifold	Nozzle valve control	EtherNet/IP
SICK Encoder / Reject Station Encoder	Encoder pulse counting module	EtherNet/IP
Baumer Camera V2.0 (GigE)	Top fill reveal camera at 12.5 m	GigE Vision
Baumer Camera PHASE V2.1 (GigE)	Finished bar reveal camera at 24.6 m	GigE Vision

7. Calibration Procedures

7.1 Phase V1 – SICK Ruler-3000 Camera Height Calibration

Reference: York Manual Section 5.4 (Engineering Values page) and Section 7.6 (Vision System Troubleshooting).

- **Tool Required:** SICK SOPAS Engineering Tool (free download) + laptop with network cable.
- **Step 1:** Connect laptop to the Beckhoff IPC network via the 'PROG NETWORK' port on the inspection cabinet. Set laptop IP to 192.168.1.x subnet.
- **Step 2:** Open SICK SOPAS. Connect to the Ruler-3000 (IP: TBD). Navigate to 'Stream Setup' to view the live 3D depth profile.
- **Step 3:** Place a reference mould (or flat calibration plate) on the conveyor directly under the laser line. Confirm the belt is stationary.
- **Step 4:** Measure the working distance shown in SOPAS. Adjust the camera mount height until WD reads 770 mm (± 5 mm tolerance).

7.2 Encoder Speed Calibration

(See Section 3.3.1 for full encoder calibration procedure)

- **Target:** Encoder pulse rate = $200 \times \text{pulses_per_mm}$ pulses/second at 0.2 m/s.
- **Tolerance:** $\pm 2\%$ of target pulse rate over a 10-second moving average.
- **Maintenance:** Re-verify encoder calibration after any conveyor belt tension adjustment, chain replacement, or sprocket re-engagement.

7.3 Phase 2 – Baumer Camera Calibration

- **Working Distance Verification:** Place a reference target (e.g., calibration grid plate 922 mm $\times$ 500 mm) on the conveyor. Verify in Baumer SDK that the full 922 mm width is visible and in focus.
- **mm/Pixel Verification:** Measure known features on the calibration plate (e.g., 100 mm spacers). Confirm pixel width = $100 \text{ mm} \div 0.174 \text{ mm/pixel} = 575 \text{ pixels}$ (± 10 pixels tolerance). Adjust lens focus/distance if outside tolerance.
- **Polarization Check:** With the polarizing filter on the lens and polarized light active: point a specular reference (mirror-finish metal plate) under the camera. The specular highlight should be extinguished (dark or very dim). Rotate the lens filter 90° — the highlight should return. Confirm cross-polarization is correctly oriented.
- **Exposure Calibration:** Set strobe pulse width to achieve a well-exposed image on a reference chocolate surface. Target: histogram peak in 100–200 DN range (out of 255) on a dark chocolate surface. Avoid saturation.

8. Data Collection & AI Model Training Plan

Phase	Activity	Owner	Duration
8.1 – Camera Installation	Install V2.0 and PHASE V2.1 cameras, lights, polarizing filters. Verify FOV, exposure, and triggering.	Faclon Labs / Neebal	Week 1–2 post-delivery
8.2 – Image Capture Mode	Run cameras in capture-only mode (no AI inference). Capture and timestamp every mould image across multiple production shifts including different chocolate recipes.	Neebal (on-site) + Faclon (remote monitoring)	3–4 weeks
8.3 – Defect Collection	Collect physical defect samples identified by Nestlé QA: air bubbles, exposed wafers, cracks. Photograph with the installed cameras. Flag images in labelling tool.	Nestlé QA Team + Faclon	Concurrent with 8.2
8.4 – Image Transfer	Transfer captured images to Faclon Labs via Secure SFTP / VPN tunnel (method to be agreed with Neebal + Nestlé IT).	Neebal IT + Faclon	Concurrent with 8.2
8.5 – Label Validation	Nestlé QA validates and approves all defect labels before training. Minimum 500 labelled images per defect class required.	Nestlé QA + Faclon	Week 3–5
8.6 – Model Training	Faclon trains Faclon Surface AI Detection model + classical pipeline on labelled dataset.	Faclon Labs	Week 6–7
8.7 – Model Deployment	Faclon deploys trained model to Beckhoff IPC via Neebal. Validate inference latency (<400 ms target).	Faclon + Neebal	Week 8
8.8 – Phase 2 Live	Phase 2 rejection activated. Initial operation under Nestlé QA supervision. Model fine-tuned based on false positives/negatives.	All parties	Week 8+

9. Defect Criteria Matrix

All stages operate under a hard **1.0 mm minimum detectable feature size** limit. Features or anomalies smaller than 1.0 mm are excluded from active rejection rules.

Defect Class	Inspection Stage	PASS Criteria	FAIL / Action
Wafer Length (Short)	Phase V1	≥80% of expected wafer length is present	REJECT: <80% wafer length present
Wafer Delamination (Height)	Phase V1	Bar with ONLY 1 wafer layer missing (height reduction <5 mm)	REJECT: 2 or more wafer layers missing (height reduction ≥5 mm)
Double-Stacked Wafer ('Sticker')	Phase V1	Wafer height within maximum acceptable range	REJECT: Wafer height exceeds maximum (two wafers stacked)
Detection Rate Alarm	Phase V1	≤10 detections per 1-minute window	ALARM: HMI alert to operator; upstream process fault suspected
Air Bubbles – Minor	V2.0 / PHASE V2.1	Air bubble diameter <3.0 mm	WARN: Log to HMI. No rejection.
Air Bubbles – Major	V2.0 / PHASE V2.1	Air bubble diameter <3.0 mm	REJECT: Air bubble ≥3.0 mm diameter
Exposed Base Wafer	V2.0 / PHASE V2.1	Exposed wafer area ≤10 mm ²	REJECT: Exposed wafer area >10 mm ²
Surface Scratches / Minor Flaws	V2.0 / PHASE V2.1	Anomaly size 1.0 mm to 3.0 mm (minor)	WARN: HMI warning, diagnostic log. No rejection.
Logo Clarity / Crack	PHASE V2.1	No cracks through finished logo	REJECT: Visible crack through bar logo

— END OF DOCUMENT —

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